Introduction to Object Oriented Programming

林致宇

Object Oriented Programming

- Object-oriented programming (OOP) is a programming paradigm based on the concept of "objects", which may contain data, in the form of fields, often known as **attributes** (**variables**); and code, in the form of procedures, often known as **methods** (**functions**).

Source: https://en.wikipedia.org/wiki/Object-oriented_programming

Example

- Car
  - Attributes (Variables)
    - speed
    - color
  - Methods (Functions)
    - run()
    - stop()

- Person
  - Attributes (Variables)
    - height
    - weight
  - Methods(Functions)
    - eat()
    - sleep()

Classes and Objects

- A class (a new data type) is a blueprint which you use to create objects (variables). An object is an instance of a class.
  - int number;
  - Car car1;

A Car Racing Game

- Classes
  - Circuit
  - Car

- Objects
  - 20 Car Objects
  - 5 Circuit Objects

Why Object Oriented?

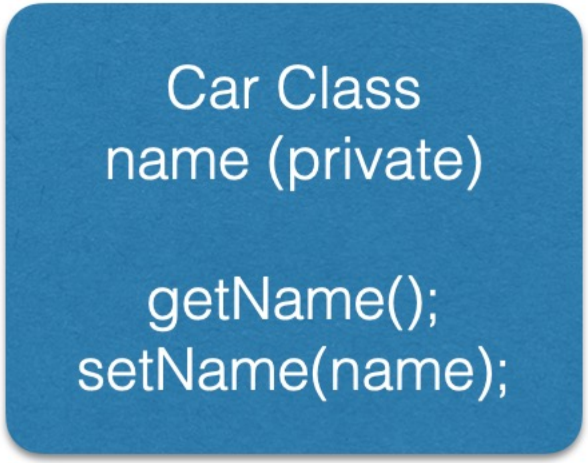
- Duplicate code is a bad thing.

- Encapsulation / Abstraction / Inheritance / Polymorphism

封裝 抽象 繼承 多形

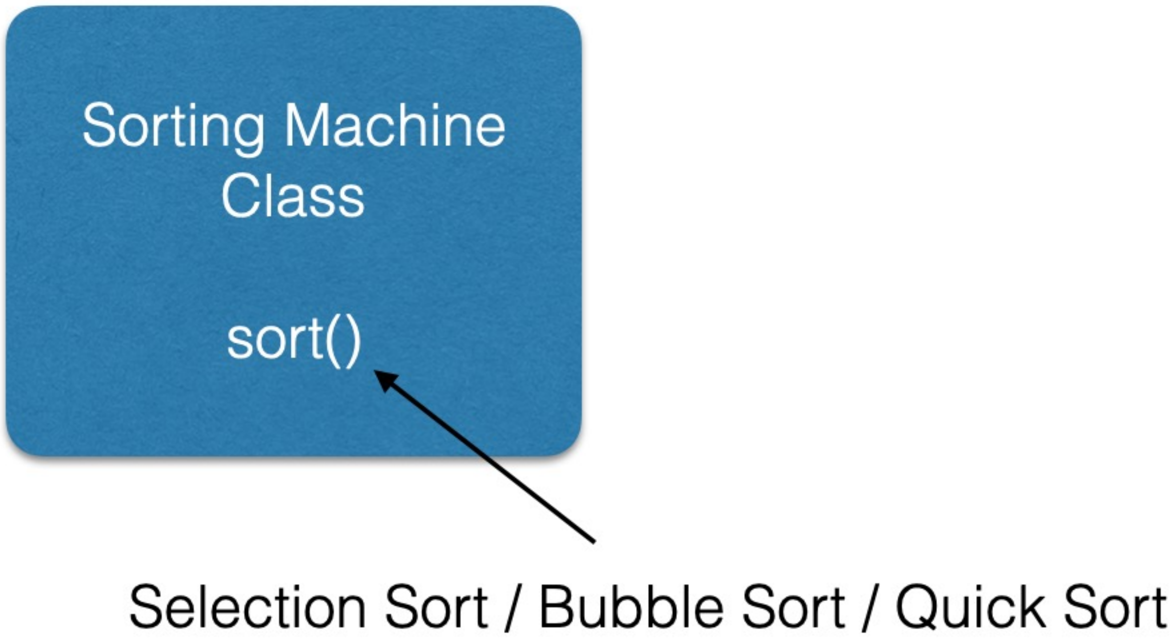
Encapsulation

- Data Hiding



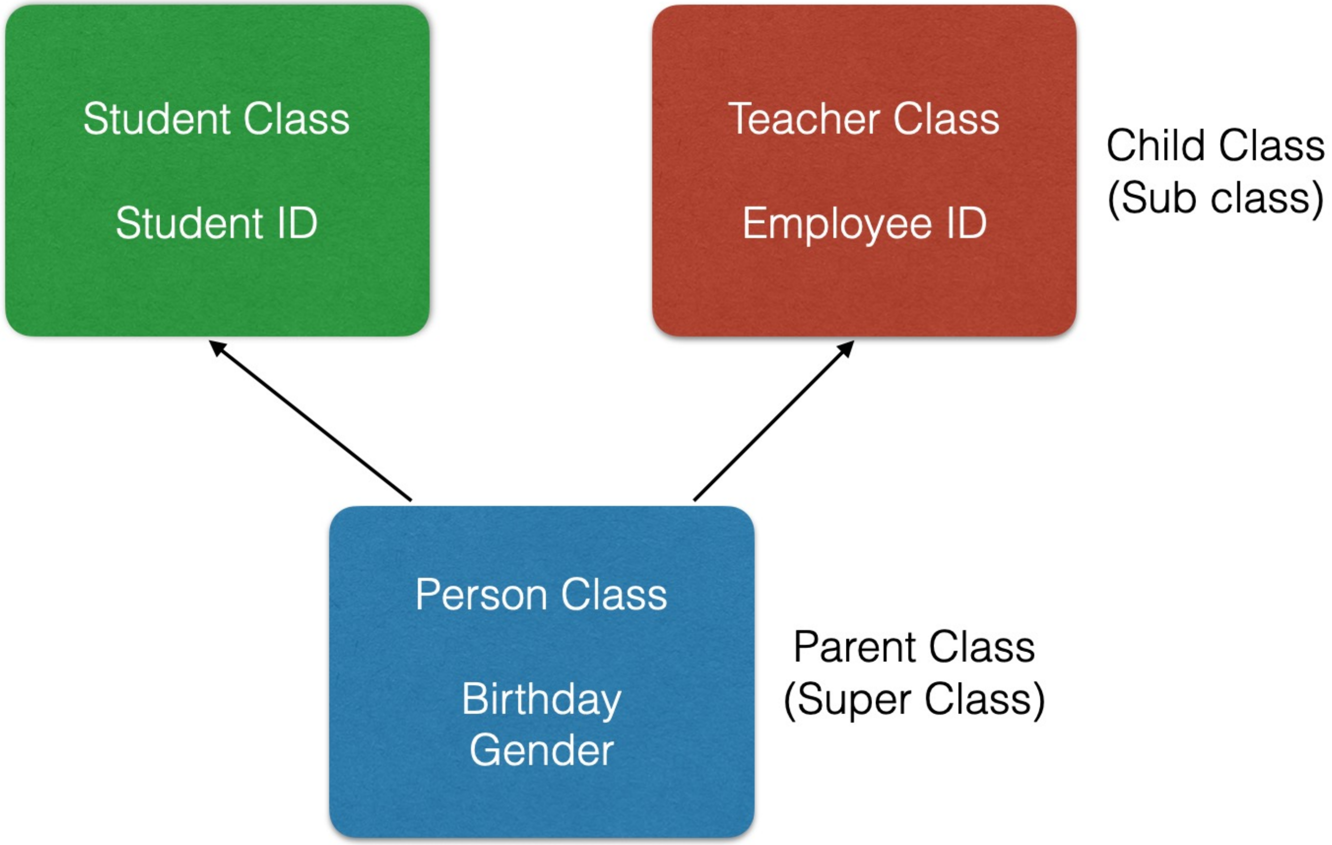
Abstraction

- Hiding the Implementation Details

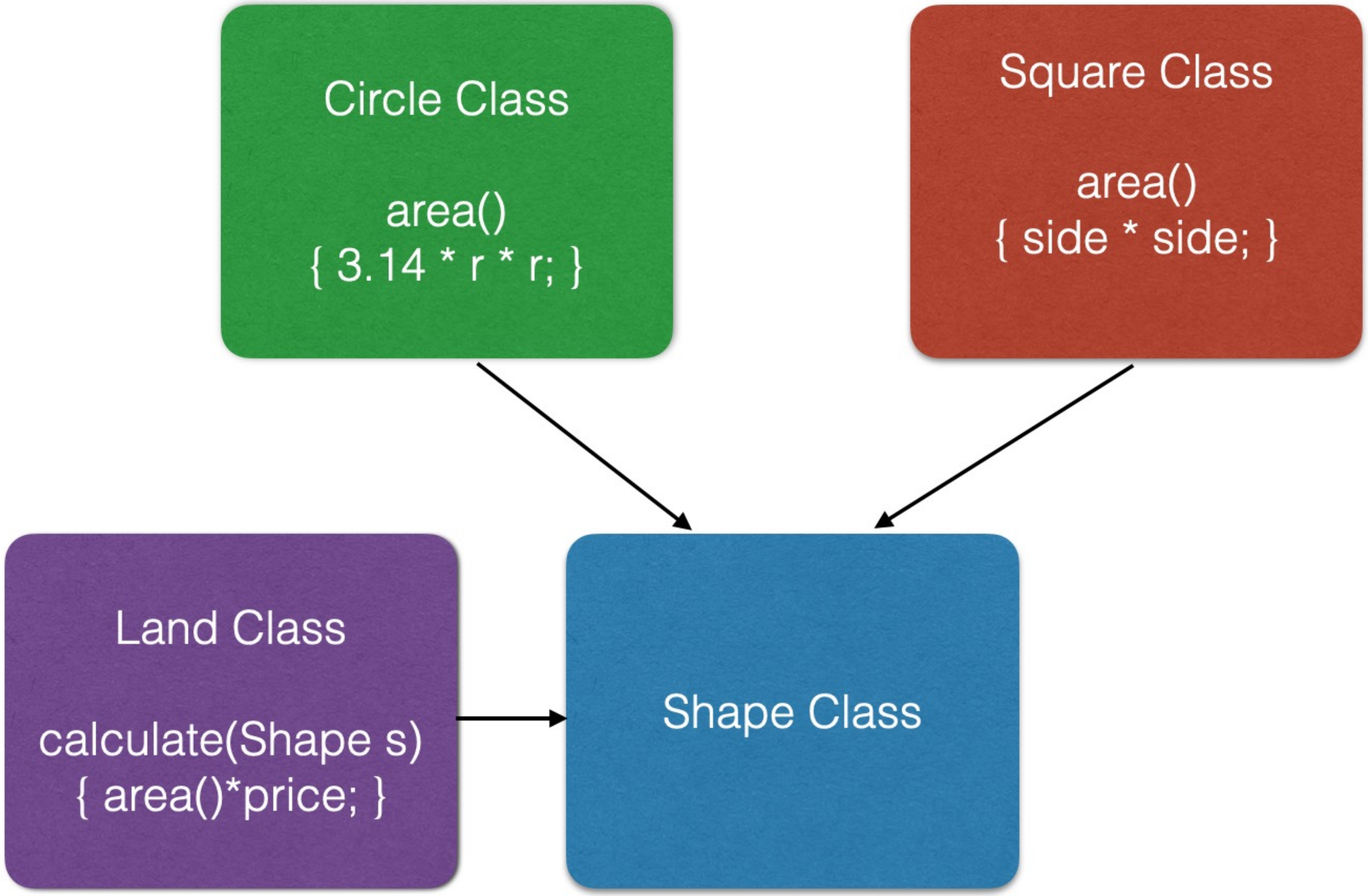


Inheritance

- Hiding the Implementation Details



Polymorphism



Suggestion

- For a beginner
  - How to create classes and objects
  - How to use objects in your program
- For an expert
  - How to use OOP features to avoid duplicate codes

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  - How to use OOP features to avoid duplicate codes